

ESP32-C3 Zero Protective Case - Prototype Tracking Sheet

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Project/team:ESP32-C3 Zero Case - Team Palozzo

| Prototype Number | Date       | Print Time (hours) | Material Use (g) | Estimated Cost (\$) | Test Result       | Comments / Issues / Improvements                                                                                                                                                                                                                                                                                                               |
|------------------|------------|--------------------|------------------|---------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prototype 1      | 30/05/2026 | 0.34               | 4.15             | \$0.12              | Fail              | The circuit board supports are too tight. The board won't fit. The pins are protruding from the case, risking damage from a fall or a short circuit when touching a conductive surface. The cover is slightly narrow, one wall remains slightly deformed after insertion. The button pins on the cover are too long, the buttons remain stuck. |
| Prototype 2      | 01/06/2026 | 0.40               | 4.82             | \$0.14              | Needs Improvement | The circuit board location has been modified to allow for easier insertion into the case. The case depth has been increased to protect the pins from damage in the event of a fall. The cover has been modified for greater precision in the housing. The button pins have been shortened.                                                     |
| Final            | 06/06/2026 | 0.35               | 4.03             | \$0.12              | Pass              | Pin-centered slots. Increased the bottom height to raise the board pins by 0.1 mm from the contact surface. Increased the number of ventilation holes and centered them. The lowest-level holes have been lowered by 1 mm so they remain below the circuit board.                                                                              |

| Summary                   |        |
|---------------------------|--------|
| Total print time (hours)  | 1.09   |
| Total material use (g)    | 13.00  |
| Total estimated cost (\$) | \$0.38 |

- Instructions
1. Enter each prototype version or final product in the table above.

2. Use the Test Result dropdown to record Pass, Fail, Needs Improvement or Retest.

3. Check that formulas calculate total print time, material use and cost before exporting to PDF.